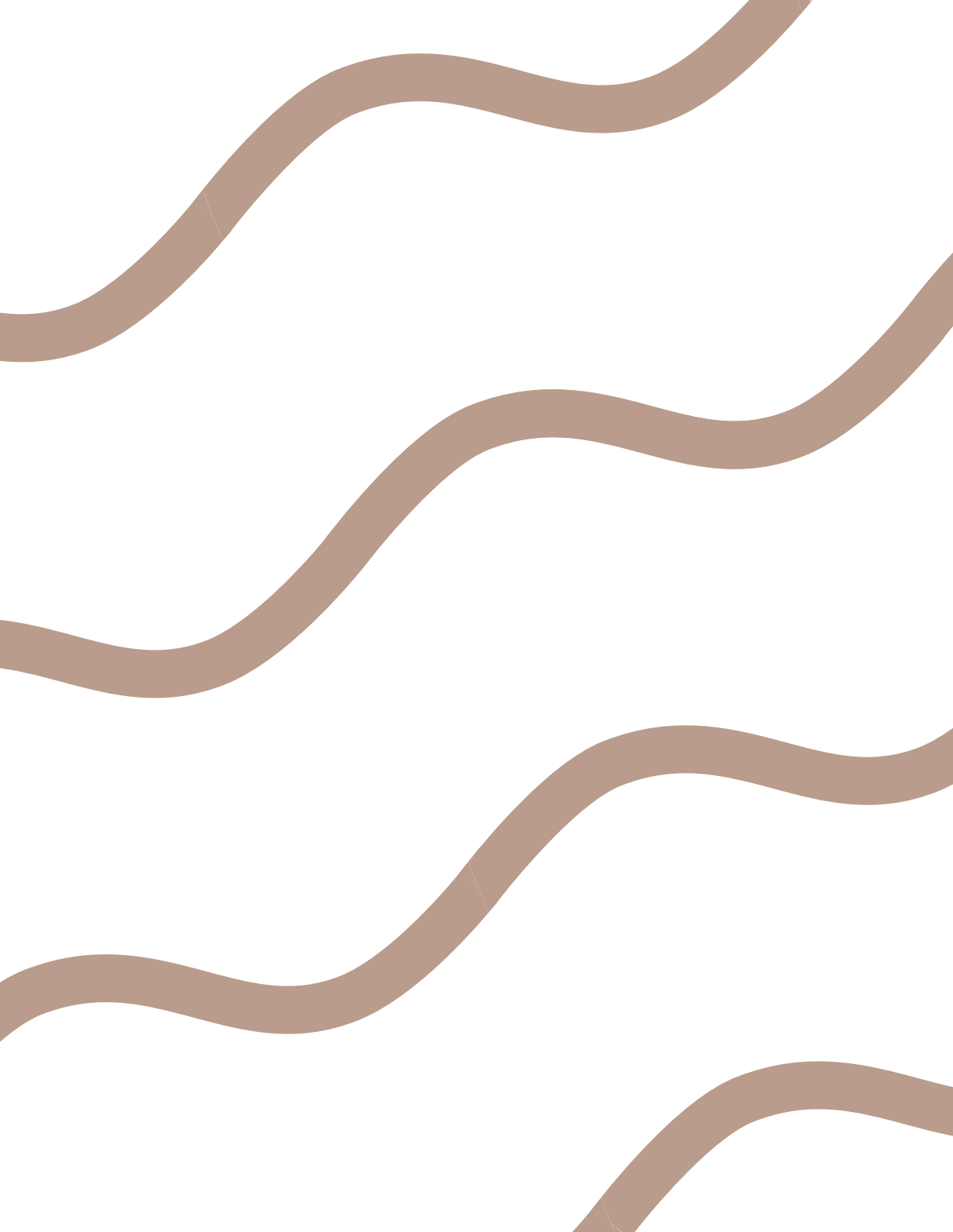




$$3. p=2 \quad n=25 \quad R^2=0.90$$

(1) 유의수준 $\alpha=0.05$ 에서 회귀모형의 유의성을 검정하라

$$H_0: \beta_1 = \beta_2 = 0$$

$$H_1: \text{적어도 하나 } \beta \neq 0$$

$$\text{자유도: } df_1 = 2 \quad df_2 = 22$$

$$F_{\text{통계량}} = \frac{\frac{0.90}{2}}{\frac{1-0.90}{25-2-1}} = \frac{0.45}{\frac{0.10}{22}} = 99$$

$$F_{\text{분포표 확인}} \quad F_{\text{임계값}} = 3.44$$

$\therefore F_{\text{통계량}}$ 이 $F_{\text{임계값}}$ 보다 크므로 귀무가설 기각

유의수준 $\alpha=0.05$ 에서 회귀모형 유의함

(2) 유의수준 $\alpha=0.05$ 에서 회귀가 유의하다고 결론내릴 수 있는 R^2 의 최솟값

- R^2 이 미지수일때 $F_{\text{통계량}}$ 이 같음

$$\frac{\frac{R_{\min}^2}{2}}{\frac{1-R_{\min}^2}{22}} \geq 3.44 \quad \text{로}$$

$$\frac{11 \cdot R_{\min}^2}{1-R_{\min}^2} \geq 3.44$$

$$11 \cdot R_{\min}^2 \geq 3.44 - 3.44 R_{\min}^2 \quad \text{전개를 하면}$$

$$14.44 \times R_{\min}^2 \geq 3.44$$

$$\therefore R_{\min}^2 \geq \frac{3.44}{14.44} \approx 0.238$$

0.238 이상이 유의하기 위한 최솟값이다.

1. $X_1, X_2, \dots, X_n$ (a) $\text{GAM}(2, \theta)$

$$f(x|\theta) = \frac{x e^{-\frac{x}{\theta}}}{\theta^2}$$

$$E(X) = 2\theta \quad \bar{X} = 2\theta$$

$$\hat{\theta}_{MLE} = \frac{\bar{X}}{2}$$

$$L(\theta|x_1, \dots, x_n) = \prod_{i=1}^n \frac{x_i e^{-\frac{x_i}{\theta}}}{\theta^2} = \frac{\prod_{i=1}^n x_i e^{-\frac{x_i}{\theta}}}{\theta^{2n}}$$

$$\log L(\theta) = \sum_{i=1}^n \log x_i - \frac{1}{\theta} \sum_{i=1}^n x_i - 2n \log \theta$$

미분하면

$$\frac{d}{d\theta} = \frac{1}{\theta^2} \sum_{i=1}^n x_i - \frac{2n}{\theta} = 0$$

$$= 2n\theta = \sum_{i=1}^n x_i \quad \therefore \theta = \frac{\bar{X}}{2}$$

$$\hat{\theta}_{MLE} = \frac{\bar{X}}{2}$$

(b) $f(x|\mu, \sigma) = \frac{1}{\sigma} e^{-\frac{x-\mu}{\sigma}} I(x > \mu)$ 지수분포, $Y = X - \mu$ 는 평균 0인 표준 지수분포

$$E(Y) = \sigma \quad \text{Var}(Y) = \sigma^2$$

$$E(X) = E(Y + \mu) = \sigma + \mu$$

$$E(X^2) = \text{Var}(X) + \{E(X)\}^2 = \sigma^2 + 2\mu\sigma + \mu^2$$

$$\mu = \bar{X} - \sigma$$

$$\bar{X} = \mu + \sigma$$

$$\bar{X}^2 = \mu^2 + 2\mu\sigma + \sigma^2 = (\bar{X} - \sigma)^2 + 2(\bar{X} - \sigma)\sigma + \sigma^2$$

$$= \bar{X}^2 - 2\bar{X}\sigma + \sigma^2 + 2\bar{X}\sigma - 2\sigma^2 + \sigma^2$$

$$= \bar{X}^2 - \sigma^2$$

$$\therefore \sigma^2 = \bar{X}^2 - \bar{X}^2$$

$$\hat{\sigma}_{MLE} = \sqrt{\bar{X}^2 - \bar{X}^2}$$

$$\hat{\mu}_{MLE} = \bar{X} - \hat{\sigma}_{MLE} = \bar{X} - \sqrt{\bar{X}^2 - \bar{X}^2}$$

$$L(\mu, \sigma|x_1, \dots, x_n) = \prod_{i=1}^n \frac{1}{\sigma} e^{-\frac{x_i - \mu}{\sigma}} I(x_i > \mu)$$

$$L(\mu, \sigma) = \sigma^{-n} e^{-\frac{\sum_{i=1}^n x_i - n\mu}{\sigma}} \prod_{i=1}^n I(x_i > \mu)$$

 μ 를 $x_{(1)}$ 로 대체하면, $\mu < x_{(1)}$ 일 때 $\mu \sim \text{MLE}$ 에 해당함

$$\hat{\mu}_{MLE} = x_{(1)} = \min(x_1, \dots, x_n)$$

$$L(x_{(1)}, \sigma) = \sigma^{-n} e^{-\frac{\sum_{i=1}^n x_i - nx_{(1)}}{\sigma}}$$

$$\ell(x_{(1)}, \sigma) = -n \log \sigma - \frac{1}{\sigma} (\sum_{i=1}^n x_i - nx_{(1)})$$

$$\frac{d}{d\sigma} = -\frac{n}{\sigma} - (\sum_{i=1}^n x_i - nx_{(1)}) \left(-\frac{1}{\sigma^2}\right) = -\frac{n}{\sigma} + \frac{\sum_{i=1}^n x_i - nx_{(1)}}{\sigma^2} = 0$$

$$-n\sigma + (\sum_{i=1}^n x_i - nx_{(1)}) = 0$$

$$\sigma = \frac{\sum_{i=1}^n x_i - nx_{(1)}}{n} = \bar{X} - x_{(1)}$$

$$\therefore \hat{\mu}_{MLE} = \min(x_1, \dots, x_n) \quad \hat{\sigma}_{MLE} = \bar{X} - x_{(1)}$$

2. $X_1, X_2, \dots, X_n \sim U(\mu - 3\sqrt{\sigma}, \mu + 3\sqrt{\sigma})$

$$E(X) = \frac{\mu - 3\sqrt{\sigma} + \mu + 3\sqrt{\sigma}}{2} = \mu \quad \text{Var}(X) = \frac{(6\sqrt{\sigma})^2}{12} = 3\sigma$$

$$\hat{\mu}_{MLE} = \bar{X} \quad 3\sigma = \bar{X}^2 - \bar{X}^2$$

$$\hat{\sigma}_{MLE} = \frac{\bar{X}^2 - \bar{X}^2}{3}$$

3. $X_1, X_2, \dots, X_n \sim \text{Poi}(\lambda) \quad g(\lambda) = \frac{1}{\lambda}$

$$P(X|\lambda) = \frac{e^{-\lambda} \lambda^x}{x!}$$

$$L(\lambda|x_1, \dots, x_n) = \prod_{i=1}^n f(x_i|\lambda) = \prod_{i=1}^n \frac{e^{-\lambda} \lambda^{x_i}}{x_i!}$$

$$L(\lambda) = \frac{e^{-n\lambda} \lambda^{\sum_{i=1}^n x_i}}{\prod_{i=1}^n x_i!}$$

$$\log L(\lambda) = \log(e^{-n\lambda}) + \log(\lambda^{\sum_{i=1}^n x_i}) - \log(\prod_{i=1}^n x_i!)$$

$$= -n\lambda + (\sum_{i=1}^n x_i) \log \lambda - \sum_{i=1}^n \log(x_i!)$$

$$\frac{d}{d\lambda} = -n + (\sum_{i=1}^n x_i) \frac{1}{\lambda} = 0$$

$$\therefore \lambda = \frac{\sum_{i=1}^n x_i}{n} \quad \hat{\lambda}_{MLE} = \frac{\sum_{i=1}^n x_i}{n} = \bar{X}$$

$$g(\hat{\lambda})_{MLE} = \frac{1}{\bar{X}}$$

Title: _____

Name: _____

Date: _____

4. $X \sim P(X=1)=\theta^2, P(X=2)=2\theta(1-\theta), P(X=3)=(1-\theta)^2$
 $(0 < \theta < 1) \quad 2, 2, 3$

$$f(x|\theta) = \begin{cases} \theta^2 & x=1 \\ 2\theta(1-\theta) & x=2 \\ (1-\theta)^2 & x=3 \\ 0 & \text{o.w.} \end{cases}$$

$$L(\theta | 2, 2, 3) = 4\theta^2(1-\theta)^4$$

$$\log L(\theta) = \log 4 + 2\log \theta + 4\log(1-\theta)$$

$$\frac{d}{d\theta} = \frac{2}{\theta} + \frac{4}{\theta-1} = 0$$

$$2(1-\theta) = 4\theta \quad \therefore \theta = \frac{1}{3}$$

5. $X_1, X_2, \dots, X_n \sim \text{Geo}(p)$

(a) $P(X=k) = (1-p)^{k-1}p$
 $E(X) = \frac{1}{p} = \bar{x}$
 $\hat{p}_{\text{mom}} = \frac{1}{\bar{x}}$

(b) $L(p | x_1, \dots, x_n) = \prod_{i=1}^n f(x_i | p) = \prod_{i=1}^n (1-p)^{x_i-1} p$
 $= p^n (1-p)^{\sum x_i - n}$

$\log L(p) = n \log p + (\sum x_i - n) \log(1-p)$

$\frac{d}{dp} = \frac{n}{p} - \frac{\sum x_i - n}{1-p} = 0$

$n(1-p) = p(\sum x_i - n) \quad p = \frac{n}{\sum x_i} = \frac{1}{\bar{x}}$
 $n - np = p \sum x_i - np \quad \therefore \hat{p}_{\text{MLE}} = \frac{1}{\bar{x}}$